

Apply Filters to SQL Queries

Project Description

The company I work for is striving to make its system more secure. It is my responsibility to ensure the system's security, investigate any potential security issues, and update employees' computers when necessary. The following steps provide examples of how I have used SQL with filters to perform security-related tasks.

Retrieve Failed Login Attempts After Business Hours

A potential security incident occurred outside of business hours (after 6:00 PM). All failed login attempts outside of business hours must be investigated.

The following code shows how I created an SQL query to filter failed login attempts outside of business hours.

```
MariaDB [organization]> SELECT *  
-> FROM log_in_attempts  
-> WHERE login_time > '18:00' AND success = FALSE;
```

event_id	username	login_date	login_time	country	ip_address	success
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0
52	cjackson	2022-05-10	22:07:07	CAN	192.168.58.57	0
69	wjaffrey	2022-05-11	19:55:15	USA	192.168.100.17	0
82	abernard	2022-05-12	23:38:46	MEX	192.168.234.49	0
87	apatel	2022-05-08	22:38:31	CANADA	192.168.132.153	0

The first section of the screenshot corresponds to my query, and the second section represents a fraction of the output. This query filters the failed login attempts that occurred after 6:00 PM. I started by selecting all data from the `log_in_attempts` table. Then, I used a `WHERE` clause combined with the `AND` operator to filter my results to only include login attempts that occurred after 6:00 PM and that were unsuccessful. The first condition is `login_time > '18:00'`, which filters for login attempts that happened after 6:00 PM. The second condition is `success = FALSE`, which filters for failed login attempts.

Retrieve Login Attempts on Specific Dates

A suspicious event occurred on 09/05/2022. Any login activity on 09/05/2022 or the day before must be investigated.

The following code shows how I created an SQL query to filter login attempts that occurred on specific dates.

```
MariaDB [organization]> SELECT *  
-> FROM log_in_attempts  
-> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1
15	lyamamot	2022-05-09	17:17:26	USA	192.168.183.51	0
24	arusso	2022-05-09	06:49:39	MEXICO	192.168.171.192	1

The first section of the screenshot corresponds to my query, and the second section represents a fraction of the output. This query returns all login attempts that occurred on 09/05/2022 or 08/05/2022. I started by selecting all data from the `log_in_attempts` table. Then, I used a `WHERE` clause combined with the `OR` operator to filter my results to only include login attempts that occurred on 09/05/2022 or 08/05/2022. The first condition is `login_date = '2022-05-09'`, which filters for logins made on 09/05/2022. The second condition is `login_date = '2022-05-08'`, which filters for logins made on 08/05/2022.

Retrieve Login Attempts Outside of Mexico

After reviewing the company's data regarding login attempts, it seems the issue lies with login attempts that occurred outside of Mexico. These login attempts need to be investigated.

The following code shows how I created an SQL query to filter login attempts that occurred outside of Mexico.

```

MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE NOT country LIKE 'MEX%';
+-----+-----+-----+-----+-----+-----+-----+
| event_id | username | login_date | login_time | country | ip_address | success |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | jrafael | 2022-05-09 | 04:56:27 | CAN | 192.168.243.140 | 1 |
| 2 | apatel | 2022-05-10 | 20:27:27 | CAN | 192.168.205.12 | 0 |
| 3 | dkot | 2022-05-09 | 06:47:41 | USA | 192.168.151.162 | 1 |
| 4 | dkot | 2022-05-08 | 02:00:39 | USA | 192.168.178.71 | 0 |
| 5 | jrafael | 2022-05-11 | 03:05:59 | CANADA | 192.168.86.232 | 0 |
| 7 | eraab | 2022-05-11 | 01:45:14 | CAN | 192.168.170.243 | 1 |
| 8 | bisles | 2022-05-08 | 01:30:17 | US | 192.168.119.173 | 0 |
| 10 | jrafael | 2022-05-12 | 09:33:19 | CANADA | 192.168.228.221 | 0 |
| 11 | sgilmore | 2022-05-11 | 10:16:29 | CANADA | 192.168.140.81 | 0 |

```

The first section of the screenshot corresponds to my query, and the second section represents a fraction of the output. This query returns all login attempts that occurred in countries other than Mexico. I started by selecting all data from the `log_in_attempts` table. Then, I used a `WHERE` clause combined with the `NOT` operator to filter countries other than Mexico. I used `LIKE` combined with `MEX%` as the matching pattern, because the dataset represents Mexico, such as `MEX` and `MEXICO`. The percentage symbol (%) matches any number of unspecified characters when used with `LIKE`.

Retrieve Marketing Department Employees

My team wants to perform updates on the computers of certain employees in the marketing department. To do this, I need to obtain information to determine which computers need to be updated.

The following code shows how I created an SQL query to filter the computers of marketing employees located in the East building.

```

MariaDB [organization]> SELECT *
-> FROM employees
-> WHERE department = 'Marketing' AND office LIKE 'East%';
+-----+-----+-----+-----+-----+
| employee_id | device_id | username | department | office |
+-----+-----+-----+-----+-----+
| 1000 | a320b137c219 | elarson | Marketing | East-170 |
| 1052 | a192b174c940 | jdarosa | Marketing | East-195 |
| 1075 | x573y883z772 | fbautist | Marketing | East-267 |
| 1088 | k865l965m233 | rgosh | Marketing | East-157 |
| 1103 | NULL | randeross | Marketing | East-460 |
| 1156 | a184b775c707 | dellery | Marketing | East-417 |
| 1163 | h679i515j339 | cwilliam | Marketing | East-216 |
+-----+-----+-----+-----+-----+
7 rows in set (0.002 sec)

```

The first section of the screenshot corresponds to my query, and the second section represents a fraction of the output. This query returns all marketing department employees located in the East building. I started by selecting all data from the `employees` table. Then, I used a `WHERE` clause combined with the `AND` operator to filter employees who work in the marketing department and are in the East building. I used `LIKE` combined with `East%` as the matching pattern because the data in the `office` column corresponds to the East building with a specific office number mentioned. The first condition is the segment `department = 'Marketing'`, which filters marketing department employees. The second condition is the segment `office LIKE 'East%'`, which filters employees located in the East building.

Retrieve Employees from the Finance or Sales Departments

The computers of employees in the finance and sales departments also need to be updated. Since a separate security update is required, I need to gather information to determine only employees working in these two departments.

The following code shows how I created an SQL query to filter the computers of employees from the finance or sales departments.

```

MariaDB [organization]> SELECT *
-> FROM employees
-> WHERE department = 'Finance' OR department = 'Sales';
+-----+-----+-----+-----+-----+
| employee_id | device_id | username | department | office |
+-----+-----+-----+-----+-----+
| 1003 | d394e816f943 | sgilmore | Finance | South-153 |
| 1007 | h174i497j413 | wjaffrey | Finance | North-406 |
| 1008 | i858j583k571 | abernard | Finance | South-170 |
| 1009 | NULL | lrodriqu | Sales | South-134 |
| 1010 | k242l212m542 | jlansky | Finance | South-109 |
| 1011 | l748m120n401 | drosas | Sales | South-292 |
| 1015 | p611q262r945 | jsoto | Finance | North-271 |
| 1017 | r550s824t230 | jclark | Finance | North-188 |
| 1018 | s310t540u653 | abellmas | Finance | North-403 |
| 1022 | w237x430y567 | arusso | Finance | West-465 |

```

The first section of the screenshot corresponds to my query, and the second section represents a fraction of the output. This query returns all employees from the finance and sales departments. I started by selecting all data from the `employees` table. Then, I used a `WHERE` clause combined with the `OR` operator to filter employees from the finance and sales departments. I used the `OR` operator rather than `AND` because I want to retrieve all employees working in either of these departments. The first condition is `department = 'Finance'`, which filters employees from the finance department. The second condition is `department = 'Sales'`, which filters employees from the sales department.

Retrieve All Employees, Except Those Working in the IT Department

My team needs to perform an additional security update on the computers of employees who are not in the IT department. To do this, I first need to gather information to determine which employees are affected.

The following example shows how I created an SQL query to filter the computers of employees who are not part of the IT department.

```

MariaDB [organization]> SELECT *
-> FROM employees
-> WHERE NOT department = 'Information Technology';
+-----+-----+-----+-----+-----+
| employee_id | device_id | username | department | office |
+-----+-----+-----+-----+-----+
| 1000 | a320b137c219 | elarson | Marketing | East-170 |
| 1001 | b239c825d303 | bmoreno | Marketing | Central-276 |
| 1002 | c116d593e558 | tshah | Human Resources | North-434 |
| 1003 | d394e816f943 | sgilmore | Finance | South-153 |
| 1004 | e218f877g788 | eraab | Human Resources | South-127 |
| 1005 | f551g340h864 | gesparza | Human Resources | South-366 |
| 1007 | h174i497j413 | wjaffrey | Finance | North-406 |
| 1008 | i858j583k571 | abernard | Finance | South-170 |
| 1009 | NULL | lrodriqu | Sales | South-134 |
| 1010 | k242l212m542 | jlansky | Finance | South-109 |
| 1011 | l748m120n401 | drosas | Sales | South-292 |

```

The first section of the screenshot corresponds to my query, and the second section represents a fraction of the output. The query returns all employees who are not part of the IT department. I started by selecting all data from the `employees` table. Then, I used a `WHERE` clause combined with the `NOT` operator to filter employees who are not part of this department.

Summary

I applied filters to SQL queries to retrieve specific information about login attempts and employee computers. I used two different tables: `log_in_attempts` and `employees`. I used the `AND`, `OR`, and `NOT` operators to filter the specific information needed for each task. I also used `LIKE` and the percentage symbol (%) to filter patterns.